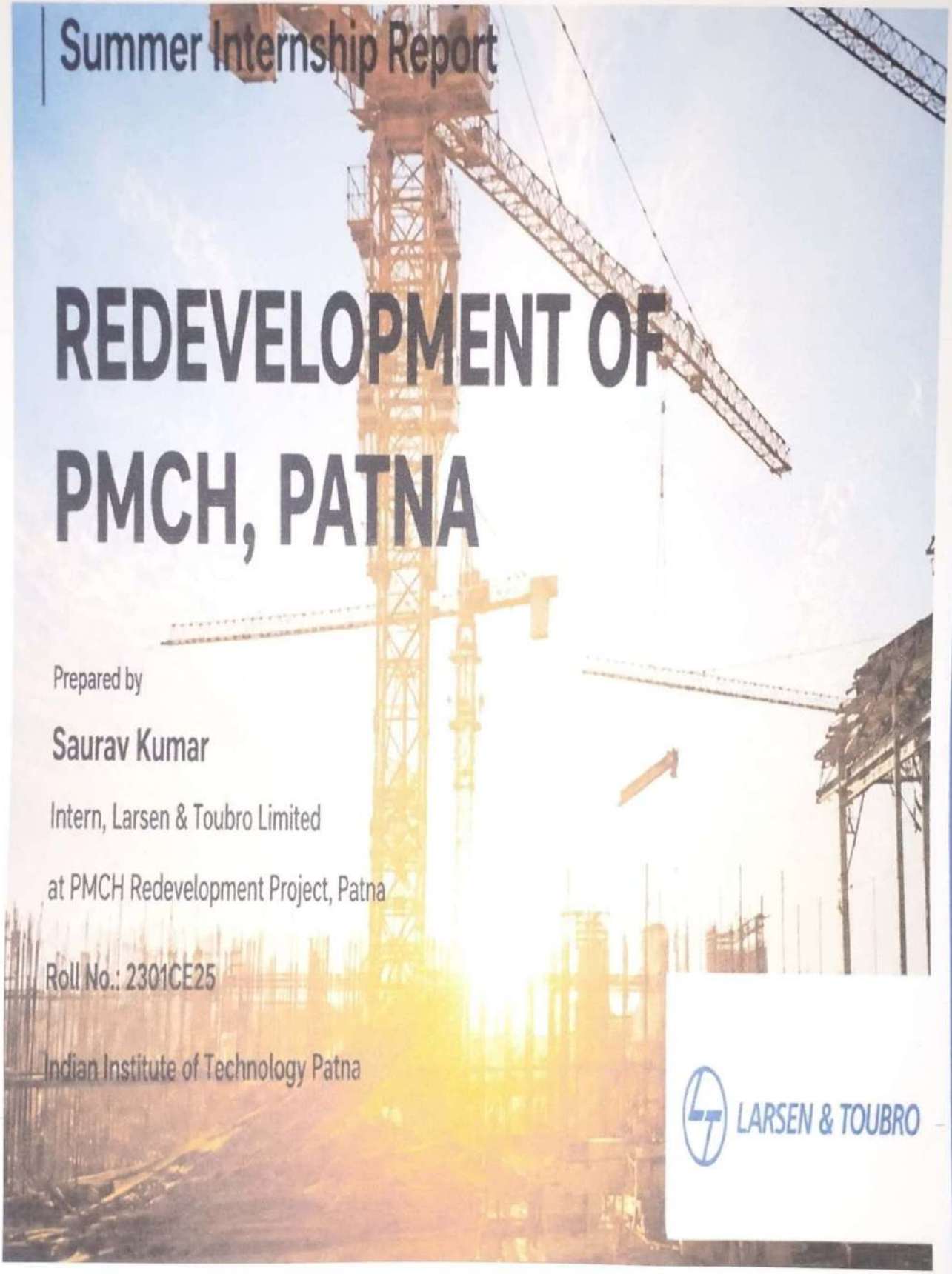


Summer Internship Report

REDEVELOPMENT OF PMCH, PATNA

Prepared by

Saurav Kumar

Intern, Larsen & Toubro Limited

at PMCH Redevelopment Project, Patna

Roll No.: 2301CE25

Indian Institute of Technology Patna

LARSEN & TOUBRO

APPROVAL

This is to certify that Mr. Saurav Kumar, Roll No. 2301CE25, has successfully completed the internship training at the Redevelopment of PMCH, Patna, under the supervision of the undersigned authorities, during the period from 15th May 2026 to 14th July 2026.

This report is submitted in partial fulfillment of the requirements for the internship program and has been reviewed and approved by the following authorities:

DESIGNATION	SIGNATURE
TFL HEAD	14/7
PROJECT DIRECTOR	13/07/26
PROJECT MANAGER	13/7/26
SITE IN-CHARGE (STRUCTURAL)	13/7/26
SITE IN-CHARGE (FINISHING)	13/7/26

SUPERVISOR'S CERTIFICATION & EVALUATION

I sincerely express my heartfelt gratitude to **Mr. Ashutosh Mishra** and **Mr. Md. Rizwan Alam** for their invaluable guidance, continuous support, and technical mentorship throughout my internship at the **Redevelopment of Patna Medical College & Hospital (PMCH), Patna** under **Larsen & Toubro (L&T) Construction**.

Their willingness to share practical knowledge, explain construction methodologies, and provide valuable insights into construction planning, execution, quality control, and site management greatly enhanced my understanding of professional civil engineering practices. Their encouragement and expert guidance made this internship an enriching learning experience and significantly contributed to my academic and professional development.

I respectfully request them to kindly evaluate my overall performance and certify the successful completion of my internship by signing below.

Particular	Supervisor – I	Supervisor – II
Name	Mr. Ashutosh Mishra	Mr. Md. Rizwan Alam
Designation	Senior Engineer	Senior Engineer
Organization	L&T Construction	L&T Construction
Project	PMCH Redevelopment Project, Patna	PMCH Redevelopment Project, Patna
Overall Evaluation (Out of 100)	96 /100	97 /100
Signature		Rizwan Alam
Date	14/7/26	13/07/2026

ABOUT THE PROJECT



Project Name:
Redevelopment of Patna Medical College and Hospital (PMCH)

Client Name:
Bihar Medical Services and Infrastructure Corporation Limited (BMSICL), Patna, Bihar

Consultant Name:
Suresh Goel and Associates, New Delhi

Contractor Name:
Larsen & Toubro (L&T)

Cost of Project:
₹6,000.00 Crore

Project Duration:
7 Years

Built-up Area:
7,48,556 sq. m.

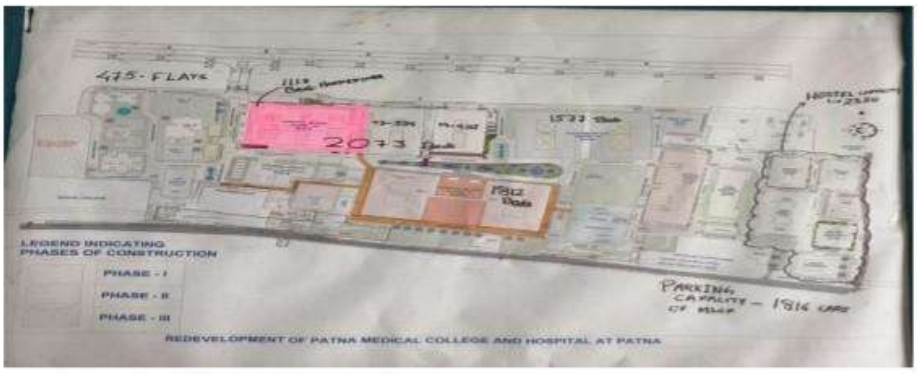
Environmental Clearance Validation:
SIA/BR/NCP/47134/2019
Dated: **29.11.2019**

PROJECT HIGHLIGHTS

- **Bed Capacity:** Upon completion, the redeveloped PMCH will have a total capacity of **5,462 beds**, including **939 Intensive Care Unit (ICU) beds**, making it one of the largest hospitals in the world.
- **Medical Education:** The redevelopment project will enhance medical education by increasing the annual **MBBS intake from 150 to 250 students** and providing **200 postgraduate (PG) seats**.
- **Smart Infrastructure:** The hospital buildings are designed using modern engineering principles. The campus will feature **green buildings**, **earthquake-resistant structures**, and **interconnected sky bridges (skyways)** to facilitate smooth movement between different blocks.
- **Connectivity:** The project is strategically connected to the **JP Ganga Path** and the proposed **double-decker corridor on Ashok Rajpath**, ensuring seamless access for ambulances, emergency vehicles, and visitors.
- **Modern Facilities:** The redeveloped campus will include advanced healthcare infrastructure such as a **pneumatic tube system** for rapid sample transportation, **modular operation theatres**, **multi-level parking facilities**, **staff residences**, and other modern support services.

CONSTRUCTION STATUS

- The redevelopment of PMCH is being carried out in **three phases** to ensure that hospital services continue without significant interruption during construction.
- A major milestone has already been achieved with the completion and inauguration of the **1117-bed Twin Tower Block**, marking significant progress in the transformation of PMCH into a modern healthcare institution.



AESTHETIC APPEARANCE









Internship Report

Redevelopment of Patna Medical College and Hospital (PMCH), Patna

Structural Works, RCC Elements, and Site Observations

Structural Works

Pile Foundation and Pile Cap Construction – Redevelopment of PMCH, Patna

A pile foundation is a type of deep foundation used to transfer the load of a structure to deeper, stronger soil or rock strata when the surface soil lacks sufficient bearing capacity. In high-rise buildings and large infrastructure projects such as the Redevelopment of Patna Medical College and Hospital (PMCH), Patna, pile foundations are essential to ensure structural stability and to minimize settlement.

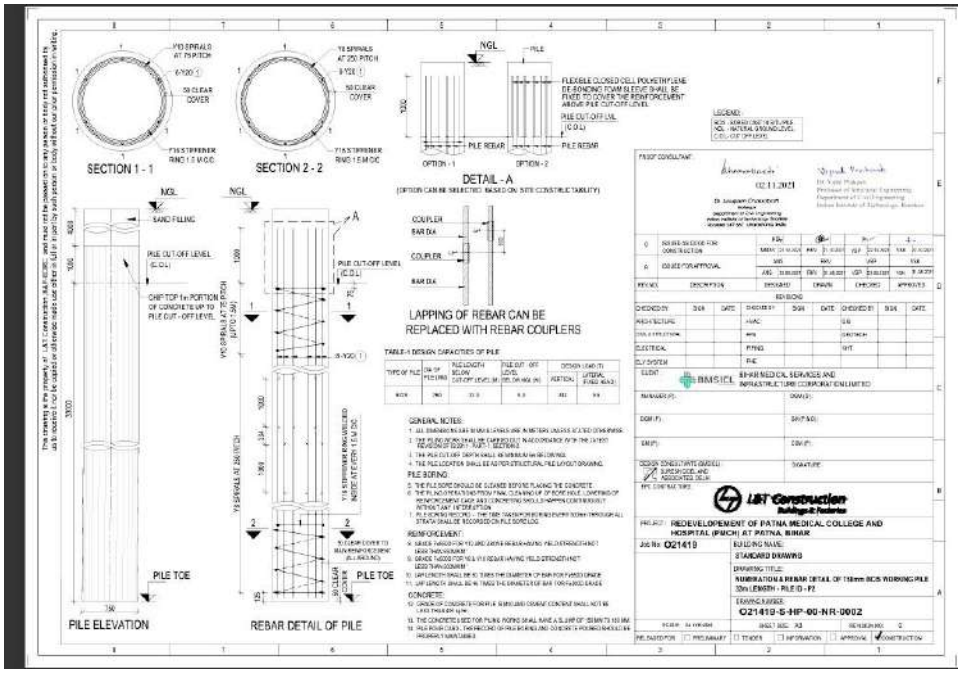
Pile Layout Drawing



Pile Layout Drawing – Hospital Block Phase-II

The figure above shows the pile layout drawing of Hospital Block Phase-II. The drawing indicates the exact location of each pile with respect to the structural grid lines. During my internship, I observed the piling work carried out at Grid No. 5, between Grid G and Grid K.

Typical Reinforcement and Elevation Details



Reinforcement and Elevation Detail of Bored Cast-in-Situ Pile

Parameter	Specification
Type of Foundation	RCC Pile
Pile Diameter	750 mm
Pile Length	33 m (as shown in the drawing)
Longitudinal Reinforcement	6 – Y20 bars
Spiral Reinforcement	Y8 @ 250 mm c/c; Y10 @ 75 mm c/c near the pile cut-off level
Concrete Cover	75mm
Concrete Grade	M30 (or as specified in the project)
Steel Grade	Fe550

Concrete Quantity Calculation for One Pile

Assuming:

- Diameter, D = 750 mm = 0.75 m
- Length, L = 33 m

Volume = $(\pi D^2 / 4) \times L$

Calculation: Volume = $(3.1416 \times 0.75^2 / 4) \times 33 = (3.1416 \times 0.5625 / 4) \times 33 = 0.4418 \times 33 = 14.58\ m^3$

Concrete required for one pile ≈ 14.6 m^3

Construction Procedure – Pile

1. Surveying and marking of pile locations.
2. Drilling / boring to the required depth using piling rigs.
3. Lowering the reinforcement cage into the borehole.
4. Installing the tremie pipe.
5. Concreting the pile using the tremie method.
6. Curing and trimming of the pile head after the concrete hardens.

Site Images



Reinforcement Cage Installation for Bored Pile



Bored Pile Drilling in Progress

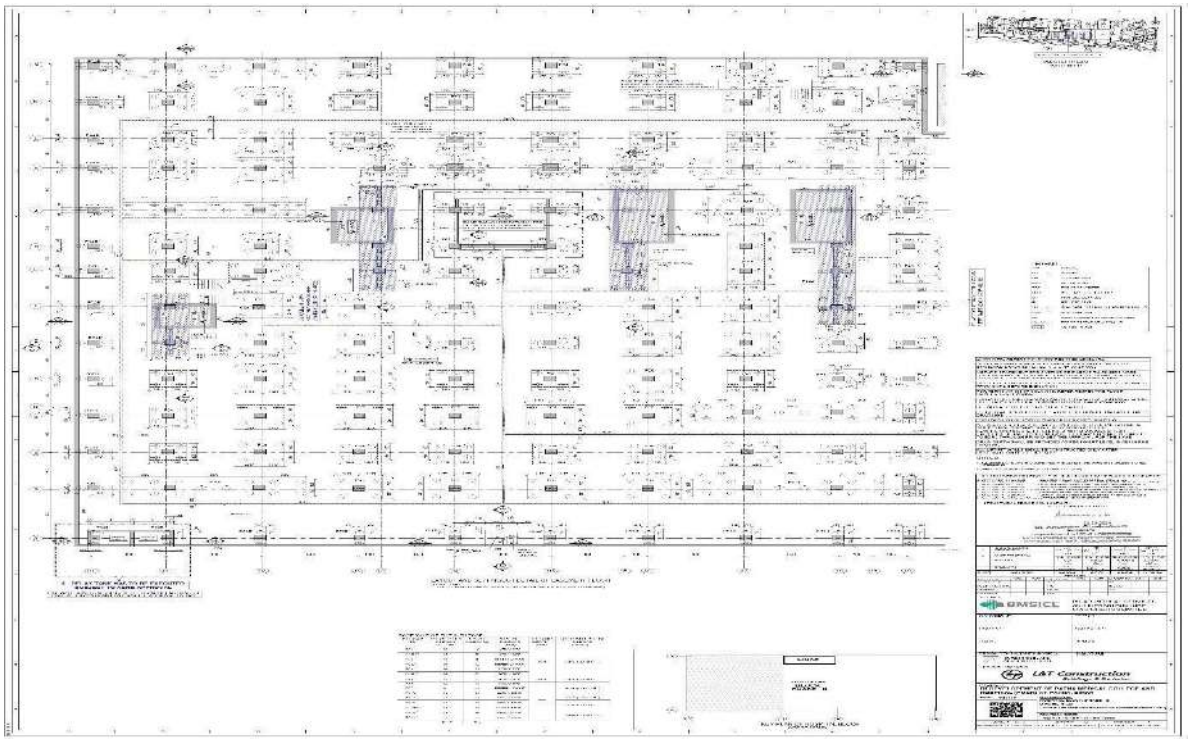


Bored Pile Concreting in Progress

Pile Cap Construction

A pile cap is a reinforced concrete structural element constructed over a group of piles to distribute the load from the column uniformly to the piles. During my internship at the PMCH Redevelopment Project, I observed the construction of Pile Cap No. PC18 at Grid No. 6, where reinforcement fixing, column starter bars, shuttering, and preparation for concreting were carried out according to the approved structural drawings.

Pile Cap Layout Drawing



Pile Cap Layout Drawing

Schedule of Pile and Pile Cap

Pile Cap ID	No. of Piles in Each Pile Cap	No. of Pile Caps	Size of Pile Cap (mm)	Pile Cap Depth (mm)	Bottom Level of Pile Cap (BOPC)
PC18	18	02	3300 × 21300	2000	-8.550 (+46.330 RL)

Construction Procedure – Pile Cap

1. Mark the pile cap location.
2. Excavate to the required depth.
3. Cut pile heads to the required level.
4. Clean the area and pile heads.
5. Fix reinforcement steel.
6. Install formwork.
7. Pour concrete.
8. Compact using a vibrator.
9. Cure the concrete.
10. Remove formwork and backfill.

Site Images – Observed Pile Cap



Fig. 1 – General View of Pile Cap



Fig. 2 – Reinforcement Fixing



Fig. 3 – Column Starter Reinforcement Embedded



Fig. 4 – Concreting of Pile Cap

Site Observation Report

Reinforced Cement Concrete (RCC) Structural Elements — Column, Beam, Slab and Staircase Construction

1. Column Construction

A column is a vertical structural member that transfers loads from slabs and beams down to the foundation. This section documents the layout, reinforcement detailing, and construction stages observed for the columns on site.

1.1 Layout Drawing

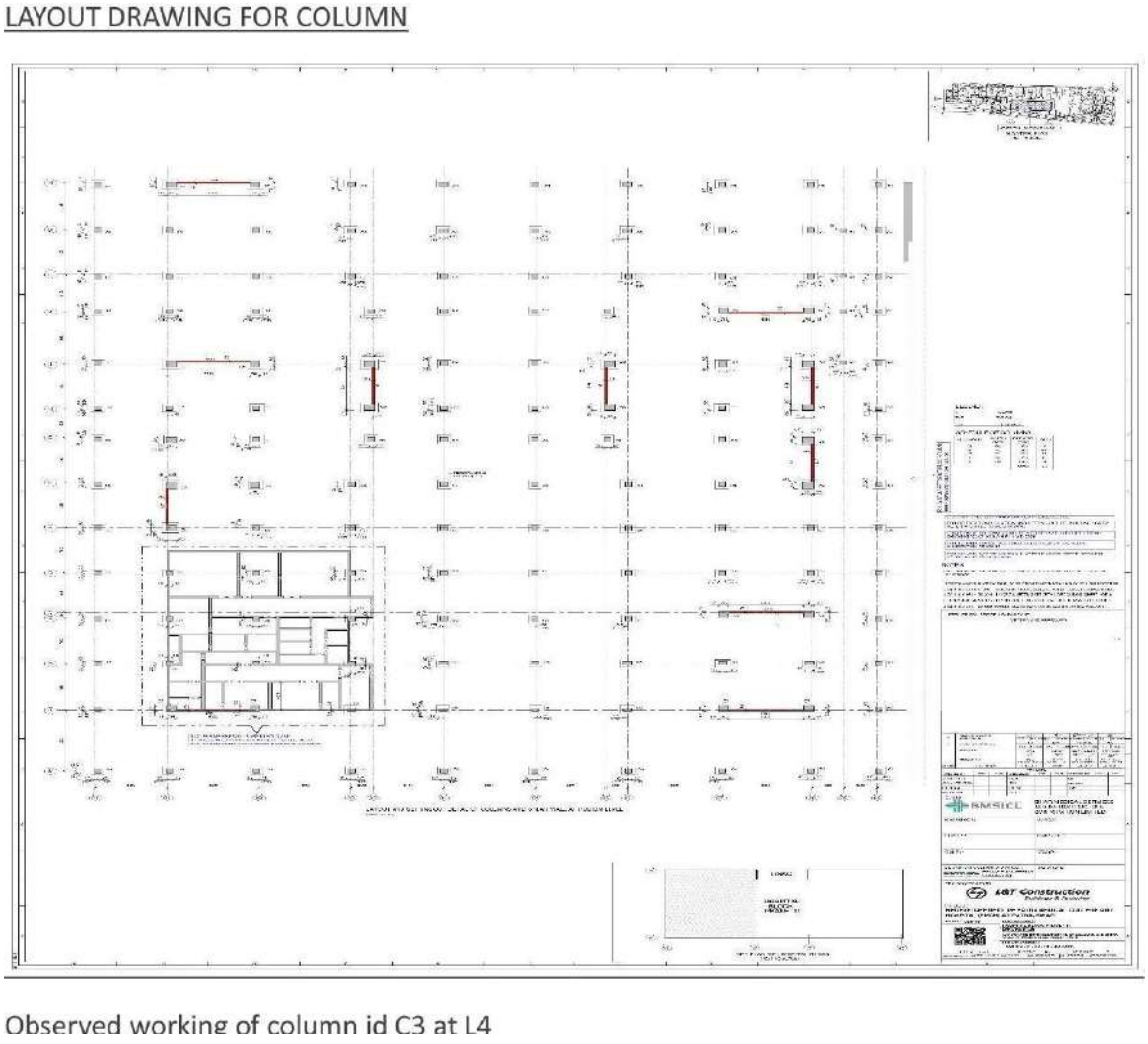


Figure 1: Column layout and seṄng-out detail at podium level.

Observed working of column ID C3 at L4.

1.2 Observed Column Detail

The working of column ID C3 at Level 4 (L4) was observed on site, with the following recorded dimensions:

Column ID	Width (mm)	Length (mm)
C3	750	600

1.3 Cross-Section and Rebar Detail

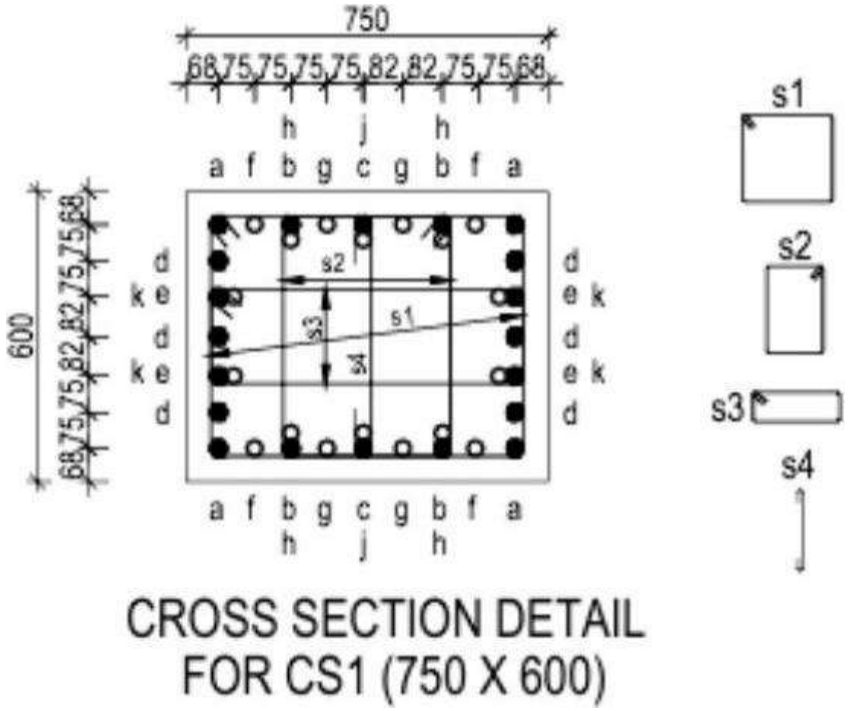


Figure 2: Cross-section detail for column type CS1 (750 mm x 600 mm).

The corresponding rebar detail for column type CS1 is summarised below:

REBAR DETAIL

1-K, 1-L, 3-M, 4-L, 4-M, 5-L, 5-M, 6-M, 8-L, 18-M, 19-L, 19-M, 20-L, 20-M, 21-M, 23-L, 23-K	750	600	CS1	TYPE-9	SECTION 1-1	4-Y25(a)+4-Y25(b')+2-Y25(c)+6-Y25(d)+4-Y25(e)	Y10 AT 200 C/C (s1) + Y8 AT 200 C/C (s2) + Y8 AT 200 C/C (s3) + Y8 AT 200 C/C (s4)	Y10 AT 150 C/C (s1) + Y8 AT 150 C/C (s2) + Y8 AT 150 C/C (s3) + Y8 AT 150 C/C (s4)
					SECTION 2-2	4-Y25(a)+4-Y25(b')+2-Y25(c)+6-Y25(d)+4-Y16(e)		
					SECTION 3-3	4-Y25(a)+4-Y20(b')+2-Y20(c)+6-Y25(d)+4-Y16(e)		
					SECTION 4-4	4-Y20(a)+4-Y20(b')+2-Y16(c)+6-Y16(d)+4-Y16(e)		
					SECTION 5-5	4-Y16(a)+4-Y16(b')+2-Y16(c)+6-Y16(d)+4-Y16(e)		
					SECTION 6-6	4-Y16(a)+4-Y16(b')+2-Y16(c)+6-Y16(d)+4-Y16(e)		

Table 1: Rebar detail for column type CS1 (750 x 600), applicable to columns 1-K, 1-L, 3-M, 4-L, 4-M, 5-L, 5-M, 6-M, 8-L, 18-M, 19-L, 19-M, 20-L, 20-M, 21-M, 23-L, 23-K.

1.4 Construction Stages

The following stages of column construction were observed on site:



Reinforcement fixing



Formwork installation



Curing



Concrete pouring

1.5 Lapping Detail

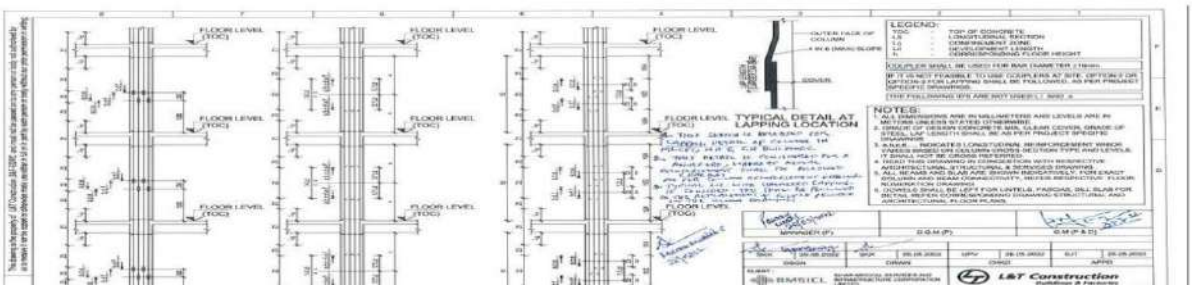


Figure 3: Typical lapping detail for column reinforcement, showing coupler, same-floor lapping, and staggered lapping options.

1.6 Construction Procedure

- Mark the column location.
- Fix the reinforcement .
- Place cover blocks.
- Install formwork.
- Check alignment and verticality.
- Pour concrete.
- Compact using a vibrator.
- Remove formwork after the required curing time.
- Cure the column.

2. Beam Construction

2.1 Activities Observed

- Observed beam reinforcement work.
- Observed beam formwork installation.
- Observed beam concreting and curing.

2.2 Construction Procedure

- Mark the beam location.
- Fix the bottom reinforcement.
- Install stirrups.
- Fix the top reinforcement.
- Place cover blocks.
- Install formwork.
- Pour concrete.
- Compact with a vibrator.
- Cure the concrete.

2.3 Layout Drawing

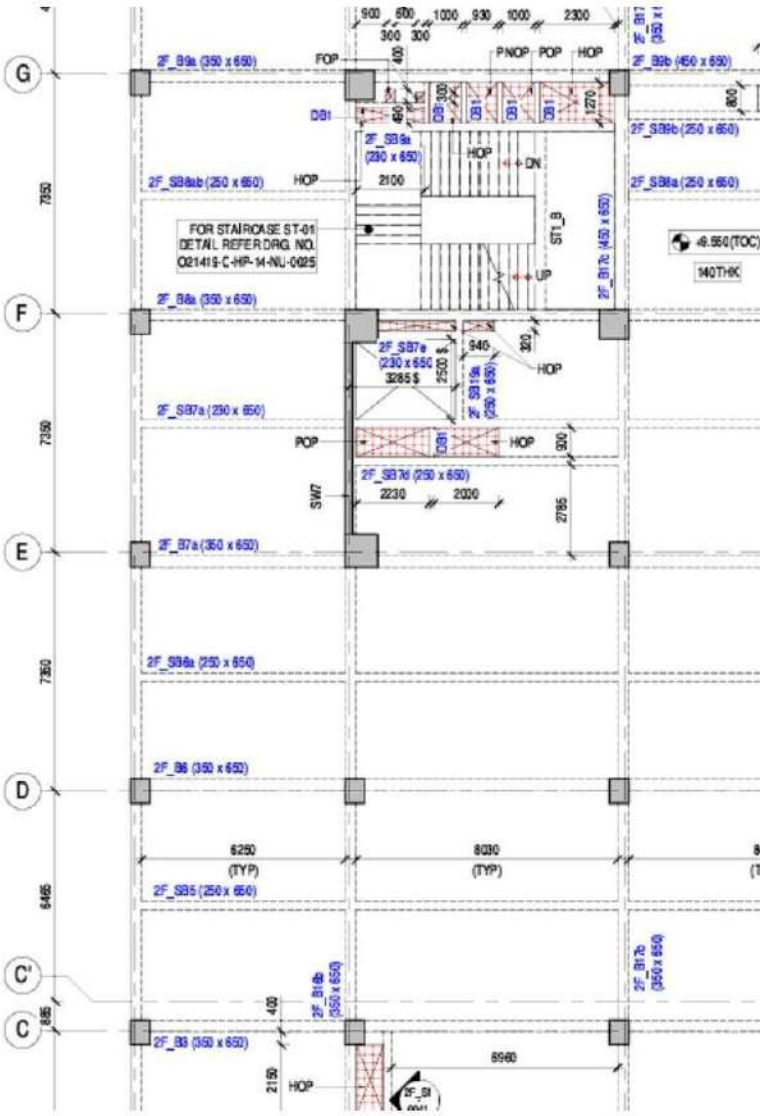


Figure 4: Observed beam and slab layout at second floor level.

Table 2: Beam reinforcement schedule (extract) corresponding to the layout above.

2.4 Site Photographs



Beam-column junction reinforcement prior to concreting.



Beam reinforcement cage supported on formwork.



Overall view of the floor showing beam and slab reinforcement in progress.

3. Slab Construction

3.1 Activities Observed

- Observed slab reinforcement work.
- Observed slab formwork installation.
- Observed slab concreting and curing.

3.2 Site Photographs



Slab concreting in progress



Slab surface aker concreting



Slab reinforcement and concreting viewed from an adjacent floor.

3.3 Reinforcement Layout

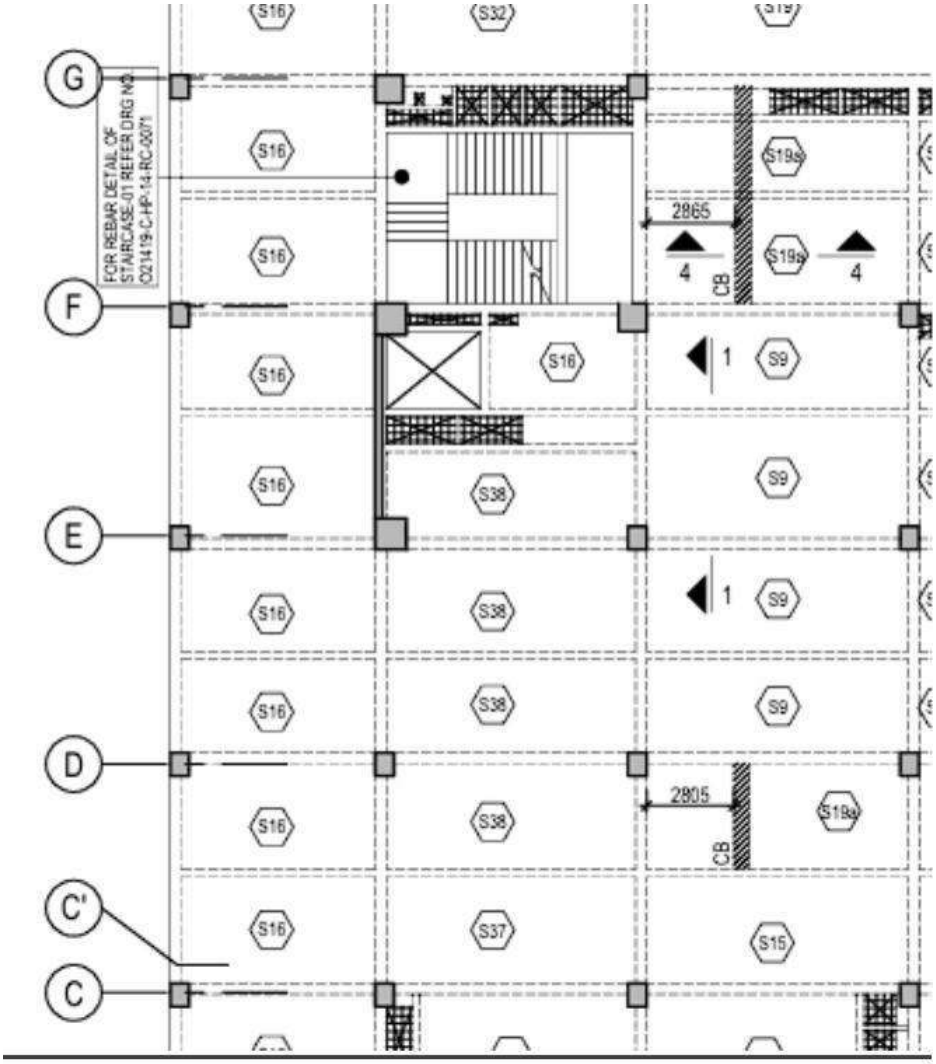


Figure 5: Slab reinforcement layout, second floor.

3.4 Rebar Schedule

The bottom and top reinforcement for each slab panel (S1 to S52 and beyond) is tabulated in the project's rebar schedule for the second floor, summarised in the drawing extract below.

REBAR SCHEDULE OF SLAB AT SECOND FLOOR

	SLAB ID	BOTTOM REBAR				TOP REBAR			
		SHORT (MAIN BAR)	SHORT (ALTERNATE BAR)	LONG (MAIN BAR)	LONG (ALTERNATE BAR)	SHORT (MAIN BAR)	SHORT (ALTERNATE BAR)	LONG (MAIN BAR)	LONG (ALTERNATE BAR)
	S1	Y8 AT 275 C/C	-	Y8 AT 300 C/C	-	Y8 AT 250 C/C	-	Y8 AT 300 C/C	-
	S2	Y8 AT 300 C/C	Y8 AT 300 C/C	Y8 AT 200 C/C	-	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 275 C/C	-
	S3	Y12 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 300 C/C	-
	S4	Y8 AT 250 C/C	Y8 AT 250 C/C	Y10 AT 250 C/C	-	Y10 AT 300 C/C	Y10 AT 300 C/C	Y8 AT 300 C/C	-
	S5	Y8 AT 125 C/C	-	Y8 AT 200 C/C	-	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 275 C/C	-
	S6	Y10 AT 125 C/C	-	Y8 AT 300 C/C	-	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-
	S7	Y8 AT 175 C/C	-	Y8 AT 200 C/C	-	Y12 AT 250 C/C	-	Y8 AT 200 C/C	-
D	S8	Y8 AT 250 C/C	Y8 AT 250 C/C	Y10 AT 150 C/C	Y10 AT 150 C/C	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 175 C/C	-
	S9	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 300 C/C	-	Y10 AT 150 C/C	Y10 AT 150 C/C	Y8 AT 300 C/C	-
	S10	Y8 AT 175 C/C	-	Y8 AT 200 C/C	-	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 225 C/C	-
	S11	Y12 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 200 C/C	-	Y12 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-
	S12	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 175 C/C	-	Y12 AT 250 C/C	-	Y8 AT 300 C/C	-
	S13	Y10 AT 125 C/C	-	Y8 AT 300 C/C	Y8 AT 300 C/C	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 200 C/C	-
	S14	Y10 AT 300 C/C	Y10 AT 300 C/C	Y8 AT 200 C/C	-	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-
	S15	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-	Y10 AT 200 C/C	Y10 AT 200 C/C	Y8 AT 300 C/C	-
	S16	Y8 AT 300 C/C	Y8 AT 300 C/C	Y8 AT 225 C/C	-	Y12 AT 250 C/C	-	Y8 AT 200 C/C	-
	S17	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 200 C/C	-	Y10 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S18	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 200 C/C	-	Y10 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 300 C/C	-
	S19	Y10 AT 200 C/C	Y10 AT 200 C/C	Y8 AT 300 C/C	-	Y12 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 300 C/C	-
	S19a	Y10 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-	Y12 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 300 C/C	-
	S20	Y10 AT 175 C/C	-	Y8 AT 175 C/C	-	Y8 AT 300 C/C	-	Y10 AT 225 C/C	-
	S21	Y20 AT 175 C/C	-	Y8 AT 200 C/C	-	Y12 AT 200 C/C	-	Y8 AT 175 C/C	-
C	S22	Y8 AT 200 C/C	-	Y8 AT 200 C/C	-	Y8 AT 175 C/C	-	Y8 AT 300 C/C	-
	S23	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 200 C/C	-	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-
	S24	Y10 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 200 C/C	-	Y10 AT 150 C/C	Y10 AT 150 C/C	Y8 AT 300 C/C	-
	S24a	Y10 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 300 C/C	-	Y10 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-
	S25	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 300 C/C	-	Y12 AT 175 C/C	-	Y8 AT 300 C/C	-
	S26	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-	Y10 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S27	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 300 C/C	-	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 300 C/C	-
	S28	Y10 AT 200 C/C	Y12 AT 200 C/C	Y8 AT 200 C/C	-	Y12 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-
	S28a	Y10 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-	Y12 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-
	S29	Y8 AT 175 C/C	-	Y8 AT 200 C/C	-	Y8 AT 250 C/C	Y8 AT 250 C/C	Y10 AT 250 C/C	-
	S30	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S31	Y8 AT 200 C/C	-	Y8 AT 200 C/C	-	Y8 AT 300 C/C	Y8 AT 300 C/C	Y8 AT 300 C/C	-
	S32	Y12 AT 200 C/C	-	Y8 AT 175 C/C	-	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 250 C/C	Y8 AT 250 C/C
	S32a	Y12 AT 200 C/C	-	Y8 AT 150 C/C	-	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 250 C/C	Y8 AT 250 C/C
B	S33	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 175 C/C	-	Y12 AT 200 C/C	-	Y8 AT 250 C/C	Y8 AT 250 C/C
	S34	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S35	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 200 C/C	-	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 300 C/C	-
	S36	Y10 AT 175 C/C	-	Y8 AT 300 C/C	Y8 AT 250 C/C	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 250 C/C
	S37	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 300 C/C	-	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S38	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 300 C/C	-	Y12 AT 250 C/C	-	Y8 AT 300 C/C	-
	S39	Y8 AT 250 C/C	Y8 AT 250 C/C	Y8 AT 225 C/C	-	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 200 C/C	-
	S40	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 200 C/C	Y8 AT 200 C/C	Y12 AT 250 C/C	Y12 AT 250 C/C	Y10 AT 250 C/C	Y10 AT 250 C/C
	S41	Y10 AT 250 C/C	Y10 AT 250 C/C	Y10 AT 250 C/C	Y10 AT 250 C/C	Y10 AT 200 C/C	Y10 AT 200 C/C	Y8 AT 300 C/C	-
	S42	Y10 AT 250 C/C	Y10 AT 250 C/C	Y12 AT 200 C/C	-	Y12 AT 250 C/C	Y12 AT 250 C/C	Y8 AT 300 C/C	-
	S43	Y16 AT 200 C/C	Y12 AT 200 C/C	Y16 AT 200 C/C	Y12 AT 200 C/C	Y12 AT 150 C/C	-	Y12 AT 150 C/C	-
	S44	Y12 AT 150 C/C	-	Y8 AT 150 C/C	-	Y12 AT 150 C/C	-	Y8 AT 150 C/C	-
	S44a	Y12 AT 150 C/C	-	Y8 AT 150 C/C	-	Y12 AT 150 C/C	-	Y8 AT 125 C/C	-
A	S45	Y10 AT 200 C/C	Y12 AT 200 C/C	Y12 AT 200 C/C	-	Y10 AT 150 C/C	Y12 AT 150 C/C	Y10 AT 250 C/C	Y10 AT 250 C/C
	S46	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 300 C/C	-	Y8 AT 200 C/C	Y8 AT 200 C/C	Y8 AT 300 C/C	-
	S47	Y10 AT 150 C/C	Y10 AT 150 C/C	Y8 AT 300 C/C	-	Y10 AT 150 C/C	Y12 AT 150 C/C	Y8 AT 300 C/C	-
	S48	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 300 C/C	-	Y8 AT 150 C/C	Y8 AT 150 C/C	Y8 AT 300 C/C	-
	S49	Y12 AT 200 C/C	Y8 AT 200 C/C	Y12 AT 200 C/C	-	Y8 AT 300 C/C	-	Y8 AT 300 C/C	-
	S50	Y10 AT 250 C/C	Y10 AT 250 C/C	Y8 AT 175 C/C	-	Y8 AT 300 C/C	Y8 AT 300 C/C	Y8 AT 300 C/C	Y8 AT 300 C/C
	S51	Y12 AT 200 C/C	-	Y12 AT 300 C/C	-	Y12 AT 300 C/C	Y12 AT 300 C/C	Y8 AT 300 C/C	-

Table 3: Rebar schedule of slab at second floor (extract).

3.5 Typical Rebar Details

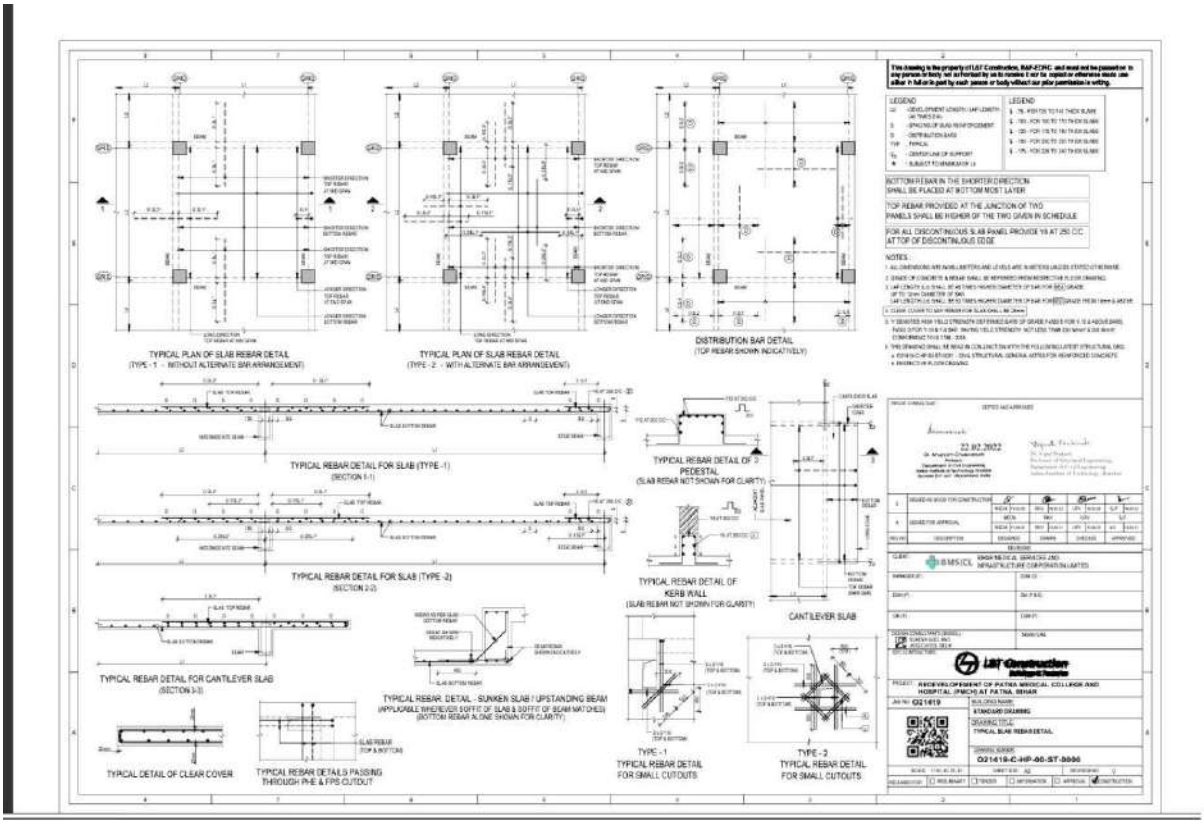


Figure 6: Typical rebar details for slab, pedestal, kerb wall, and cantilever slab sections.

4. Staircase Construction

A staircase is a structural component that provides safe movement between different floors of a building. In RCC buildings, it is constructed using reinforced concrete, with reinforcement bars, formwork, and concrete placed as per the structural drawings.

4.1 Construction Procedure

- Install staircase formwork.
- Fix reinforcement as per the drawing.
- Place cover blocks.
- Check dimensions and reinforcement.
- Pour concrete.
- Compact using a vibrator.
- Cure the concrete.

4.2 Observations

- Reinforcement was fixed according to the structural drawing.
- Cover blocks maintained the required concrete cover.
- Formwork was properly supported before concreting.
- Concrete was compacted and cured after casting.

4.3 Layout Drawings

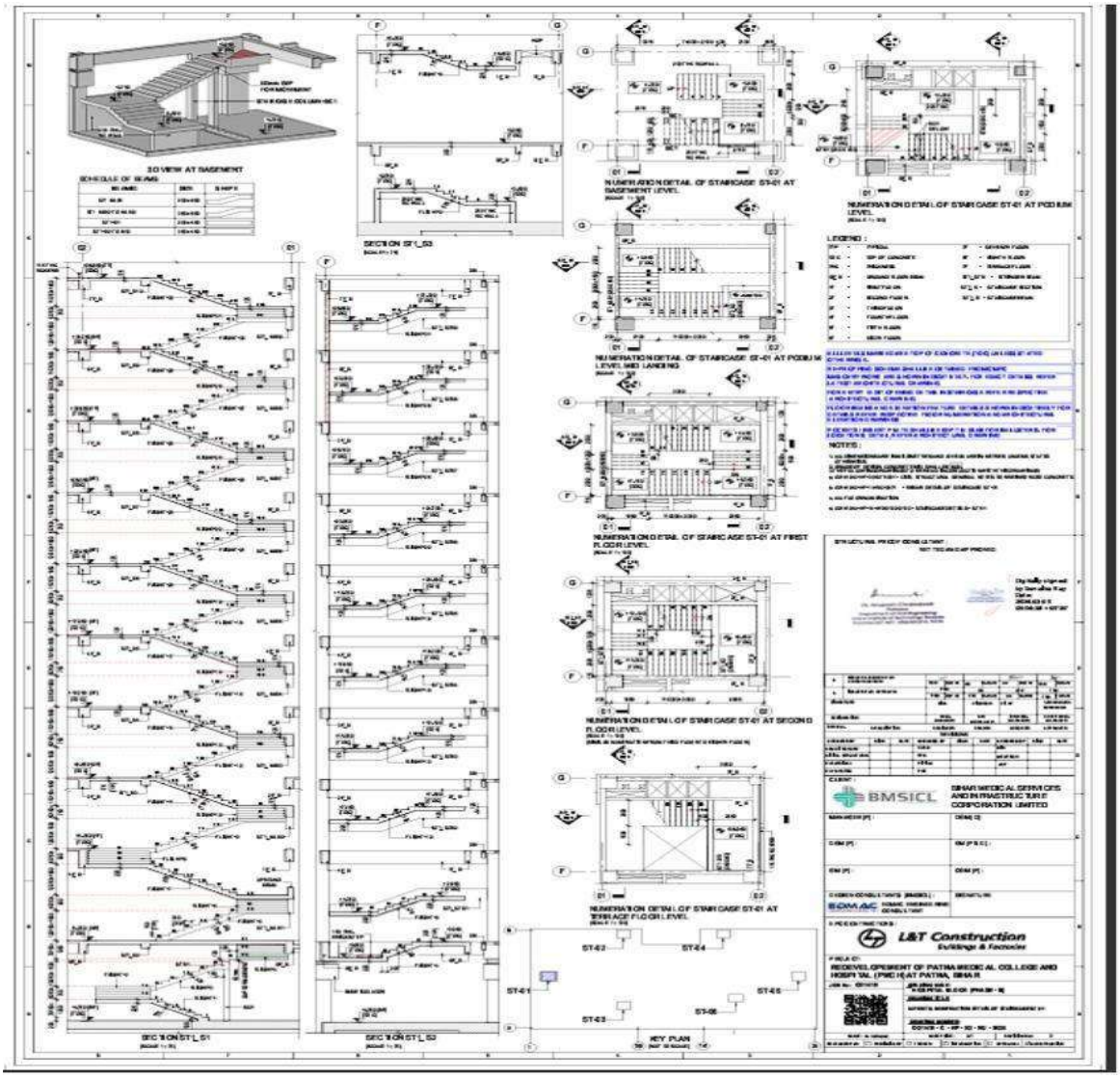


Figure 7a: Staircase layout, sections, and reinforcement detail.

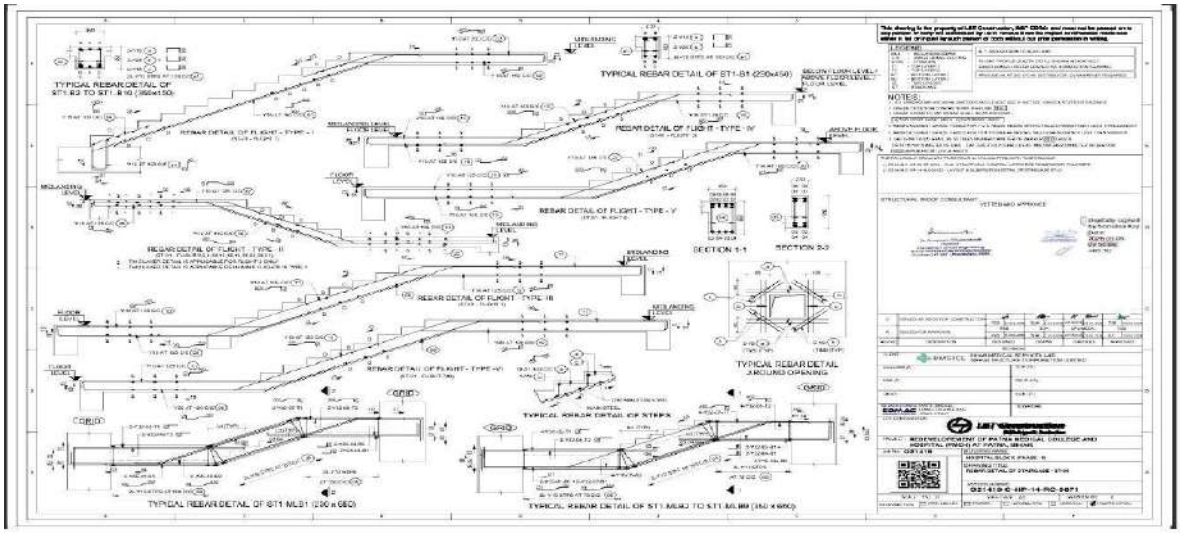


Figure 7b: Typical rebar detail for staircase flights and steps.

4.4 Site Photographs



Staircase reinforcement fixed prior to concreting



Staircase formwork installed for concreting

1. Introduction to Finishing Works

Construction finishing works represent the final stage of a building project. These activities enhance the appearance, durability, and functionality of the structure, transforming a completed structural shell into a usable, aesthetically pleasing facility. During my internship, I had the opportunity to observe and actively participate in a range of finishing activities, including brickwork, plastering, flooring, false ceiling installation, painting, quality inspection, and project handover.

Sequence of Finishing Activities

The finishing works on site generally followed the sequence shown below:

- Brick Work
- MEP Clearance (verification of mechanical, electrical and plumbing works)
- Plastering
- Flooring
- False Ceiling
- Painting
- Final Cleaning
- Quality Inspection
- Project Handover

2. Brick Work

Brick work is the process of constructing walls and other building components by laying bricks in a specified bond and binding them together with mortar.

2.1 Objective

The main objectives of brick work are:

- To provide strength and stability to the building.
- To construct internal and external walls in accordance with the approved drawings.
- To ensure proper alignment, level, and durability of the masonry.
- To provide a suitable surface for plastering and other finishing works.

2.2 Materials Used

- Fly ash bricks (230 × 115 × 75 mm)
- Ordinary Portland Cement (OPC)
- Fine sand
- Clean water
- Reinforcement bars (8 mm diameter, where required)



Fly ash bricks



Cement



Sand

2.3 Tools and Equipment

- Trowel
- Hammer
- Plumb bob
- Measuring tape
- String line
- Wheelbarrow
- Mortar pan
- Measuring box (300 × 300 × 275 mm)
- Shovel



Wheelbarrow



Shovel



Hammer



Measuring box



Trowel

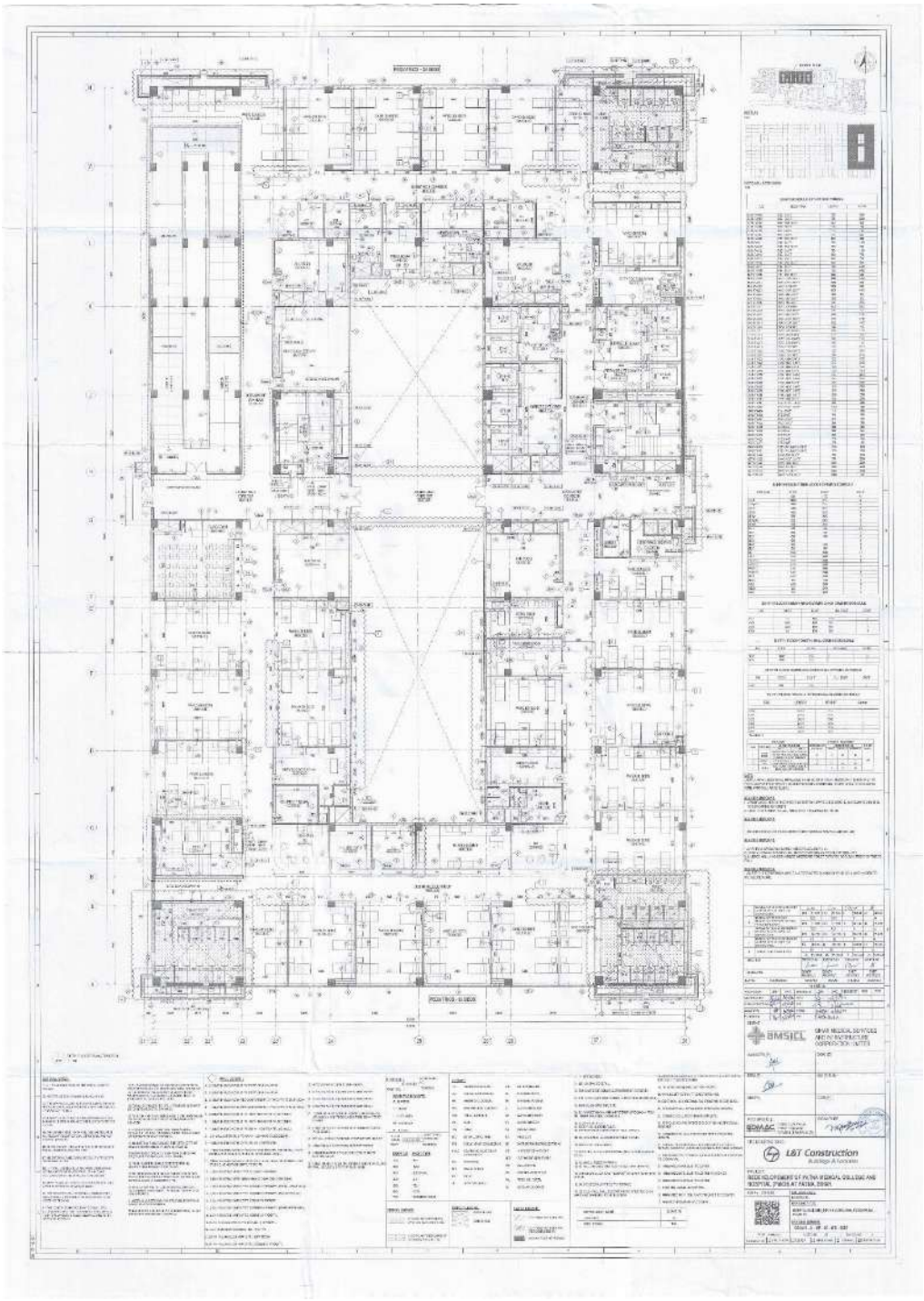
2.4 Procedure

1. The wall layout was marked according to the approved architectural drawings.
2. Bricks were soaked in water before laying to improve the bond with mortar.
3. Cement mortar was prepared in the specified mix ratio.
4. A mortar bed was spread uniformly, and bricks were laid in the required bond pattern.
5. The thickness of mortar joints was maintained uniformly throughout the work.
6. Wall alignment, level, and plumb were checked continuously using a spirit level and plumb bob.
7. Reinforcement bars were provided where specified in the drawings.
8. The completed brickwork was cured with water for the required duration to achieve the desired strength.

2.5 Key Site Observations

1. Mortar Mix Ratio: A cement-sand mortar mix of 1:4 was used for 115 mm thick brick walls, while a 1:6 mortar mix was used for 230 mm thick brick walls, as per the project specifications.
2. Horizontal Reinforcement: In 115 mm thick brick walls, two 8 mm diameter reinforcement bars were provided horizontally after every four courses (layers) of brickwork to improve wall stability and resistance to cracking.
3. Mullion Provision: For brick walls exceeding 4 m in length, mullions were provided at suitable intervals to increase stiffness and minimise the risk of cracking due to the wall's slenderness.
4. Lintel Level: Lintel beams were provided above door and window openings at a height of approximately 2100 mm to 2400 mm from the finished floor level (FFL), depending on the architectural drawings and opening dimensions.
5. Labour Productivity: A team consisting of one mason and one helper was capable of completing approximately 4 m^2 of 115 mm thick brickwork per day. For 230 mm thick brick walls, the same team could complete approximately 1.5 m^3 of brickwork per day, depending on site conditions and the complexity of the work.

2.6 Layout Drawing for Brick Work



Approved architectural layout drawing used for brickwork marking

2.7 Approximate Material Estimation for Brick Work

Given:

- Quantity of brickwork observed = 200 m^3
- Approximate number of bricks required for 1 m^3 of brickwork = 450 bricks

Calculation of Number of Bricks

Number of Bricks = Quantity of Brickwork × Number of Bricks per m³

= 200 × 450 = 90,000 bricks

Therefore, the approximate number of bricks required = 90,000 Nos.

Calculation of Cement Requirement (1:6 Mortar)

Given:

- Quantity of brickwork = 200 m³
- Mortar required for 1 m³ of brickwork = 30% of brickwork volume
- Dry volume factor for mortar = 1.33
- Mortar mix = 1 : 6 (Cement : Sand)
- Volume of one cement bag = 0.035 m³

Step 1 — Wet volume of mortar = 30% × 200 = 60 m³

Step 2 — Dry volume of mortar = 60 × 1.33 = 79.8 m³

Step 3 — Volume of cement = (1/7) × 79.8 = 11.4 m³

Step 4 — Number of cement bags = Cement volume ÷ Volume of one bag = 11.4 ÷ 0.035 = 325.7 ≈ 326 bags

Result: The approximate cement required for 200 m³ of brickwork using 1:6 mortar is 326 bags.

2.8 Site Images



Brickwork in progress on scaffolding



Completed brick wall



Mason laying bricks with mortar

2.9 Quality Inspection

1. The brick layout was checked before starting the brickwork to ensure compliance with the approved drawings.
2. The alignment, level, and verticality (plumb) of the walls were checked during and after construction using a plumb bob and mason's line.
3. The location and dimensions of door and window openings were verified according to the approved drawings before placing lintels.

2.10 Safety Precautions

1. Workers wore personal protective equipment (PPE), including safety helmets, safety shoes, gloves, and reflective jackets.
2. Scaffolding was inspected before use to ensure safe working conditions (a safety belt was also used above a height of 4 m).
3. Bricks and mortar were handled carefully to prevent injuries and material damage.
4. The work area was kept clean to minimise slip and trip hazards.

2.11 Learning Outcome

Through this activity, I gained practical knowledge of brick masonry construction, including wall layout, mortar preparation, bond patterns, reinforcement detailing, quality inspection, productivity assessment, and safety practices. I also learned the importance of maintaining proper alignment, level, and workmanship in order to achieve durable and high-quality brickwork.

3. Plastering

3.1 Definition

Plastering is the process of applying a layer of cement mortar to brick or concrete surfaces to obtain a smooth, even, and durable finish while protecting the surface from weathering and moisture.

3.2 Purpose

- To provide a smooth and level surface for painting and other finishes.
- To improve the appearance and durability of the building.

3.3 Materials Used

- Ordinary Portland Cement (OPC)
- Fine sand
- Clean water
- Cement mortar

3.4 Tools and Equipment

- Plastering trowel
- Aluminium straight edge
- Measuring tape
- Mortar pan

3.5 Procedure

1. Before starting plastering, it was ensured that all concealed electrical works, including conduits and electrical boxes, had been completed.
2. All chases, grooves, and openings made for electrical and plumbing services were filled with cement mortar.
3. Chicken mesh (GI wire mesh) was fixed at the junction of brick masonry and RCC members, as well as over repaired chases, to minimise the formation of cracks in the plaster.
4. Plaster dots (bull marks) of 12–15 mm thickness were provided at suitable intervals to maintain a uniform plaster thickness and achieve a level surface.
5. The wall surface was cleaned to remove dust, loose particles, and excess mortar.
6. The wall surface was thoroughly wetted with water before plastering to improve the bond between the mortar and the wall.
7. Cement mortar was prepared according to the specified mix ratio and applied uniformly using a trowel.
8. The plastered surface was levelled using an aluminium straight edge, matching it with the plaster dots (bull marks) to obtain a smooth and even finish.
9. The completed plaster was cured with water for the specified period to achieve the required strength and durability.

3.6 Site Observations

1. All concealed electrical and plumbing works were completed before plastering commenced.
2. Chicken mesh was provided at the junction of RCC members and brick masonry to prevent cracks.
3. Plaster dots (bull marks) of 12–15 mm were used to maintain uniform plaster thickness.
4. The wall surface was thoroughly wetted before applying plaster.
5. Proper curing was carried out after plastering to improve strength and reduce shrinkage cracks.
6. The mortar ratio used was 1:5 for internal plaster and 1:6 for external plaster; a richer 1:3 ratio was used in areas without a false ceiling.



Applying plaster over chicken mesh



Bull marks before plaster

3.7 Quality Inspection

1. The plaster thickness was checked to ensure it met the specified requirement.
2. The surface was inspected for smoothness, evenness, and proper finishing using an aluminium straight edge.
3. Verticality and level were checked to ensure a uniform finish.
4. The plastered surface was inspected for cracks, hollowness, and debonding before acceptance.

3.8 Safety Precautions

1. Workers wore PPE, including helmets, safety shoes, gloves, and reflective jackets.
2. Scaffolding was inspected before use to ensure safe working conditions.
3. The work area was kept clean to prevent slips, trips, and falls.
4. Safe working procedures were followed while carrying out plastering at height.

3.9 Site Images



Freshly plastered internal wall



Plastered wall awaiting curing

3.10 Learning Outcome

During my internship, I gained practical knowledge of surface preparation, plaster application, plaster thickness control using bull marks, crack prevention using chicken mesh, quality inspection, and the safety practices followed during plastering works.

4. Flooring

Flooring is the process of laying floor finishing materials, such as vitrified or ceramic tiles, on a prepared base to provide a durable, level, and aesthetically pleasing surface.

4.1 Purpose

- To provide a smooth, durable, and easy-to-maintain floor surface.
- To enhance the appearance and functionality of the building.

4.2 Materials Used

- Vitrified / Ceramic tiles
- Tile adhesive or cement mortar
- Cement
- Sand
- Grout
- Clean water



Tile grout



Tile adhesive



Vitrified tiles

4.3 Tools and Equipment

- Tile cutter
- Rubber mallet
- Trowel
- Measuring tape



Hand grinder used for surface hacking

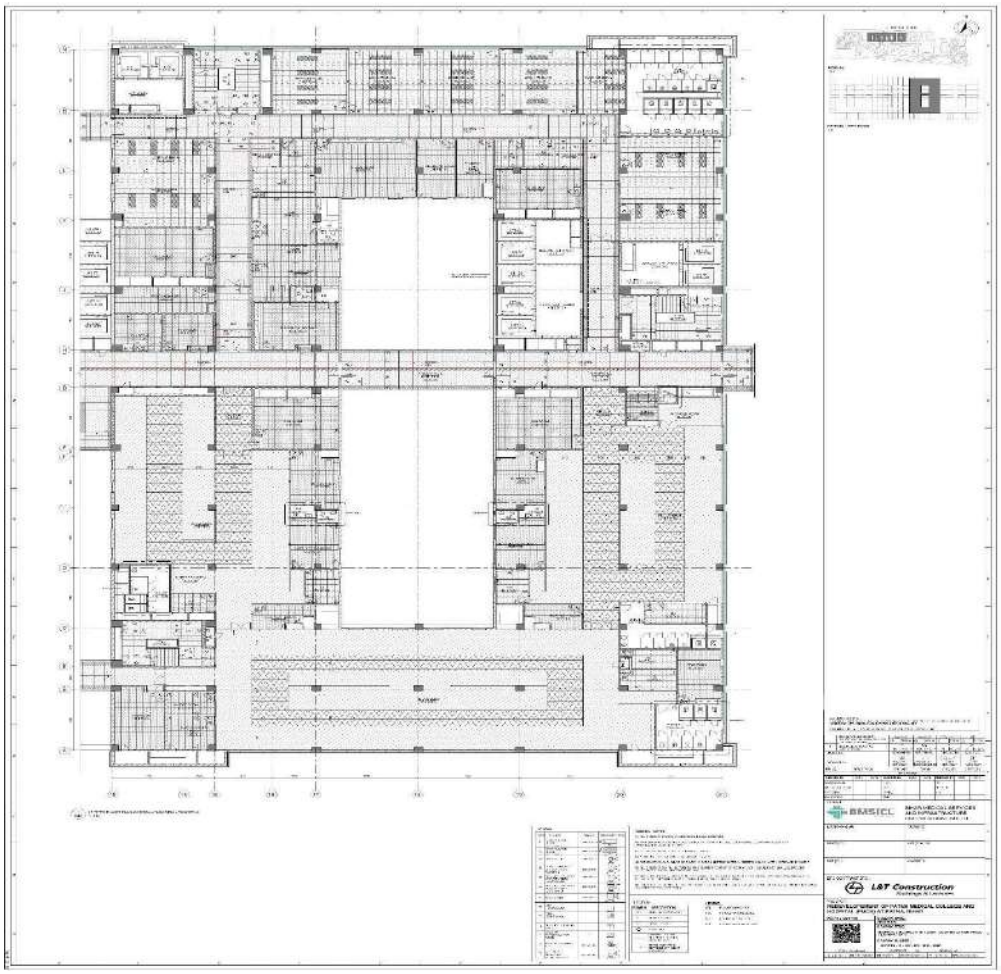


Tile cuŃng machine

4.4 Procedure

1. The floor surface was cleaned thoroughly to remove dust, loose particles, and construction debris.
2. Hacking was carried out on the concrete floor surface to create a rough texture and improve the bond between the base and the flooring material.
3. Reference levels (bull marks) were established according to the approved drawings to maintain the required floor level and slope.
4. Flooring work was started from the designated starting point specified in the approved drawings and proceeded systematically to ensure proper alignment and layout.
5. Tile adhesive or cement mortar was spread uniformly over the prepared surface.
6. Tiles were laid according to the specified pattern, maintaining the required joint spacing.
7. The level and alignment of the tiles were checked continuously using a plumb bob.
8. After the tiles were properly set, the joints were filled with grout.
9. The completed flooring was cleaned and protected until the adhesive and grout had fully cured.

4.5 Layout Drawing



Approved floor layout drawing

4.6 Site Images



Skirting work in progress

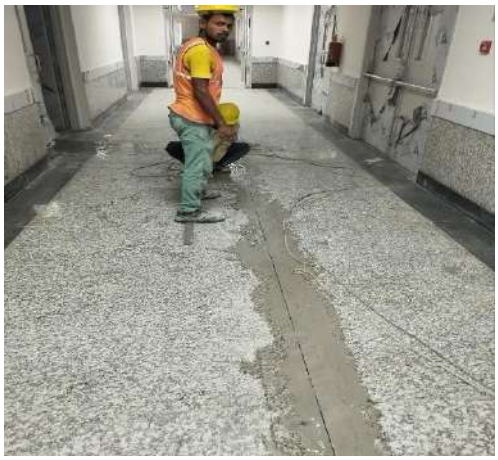


Tile work in progress

Skirting work in progress



Hacking of the concrete surface



CuŃng Granite to size

4.7 Quality Inspection

1. The floor level and tile alignment were checked.
2. Joint spacing and grout filling were inspected.
3. Tiles were checked for hollowness by tapping with a rubber mallet.
4. Damaged, chipped, or cracked tiles were replaced.

4.8 Safety Precautions

1. Workers wore PPE, including helmets, safety shoes, gloves, and reflective jackets.
2. The work area was kept clean to avoid slips and trips.
3. Tile cutting was carried out using appropriate tools and safety goggles.
4. Proper lifting techniques were followed while handling tile boxes.

4.9 Learning Outcome

During my internship, I observed the complete flooring process, including surface preparation, tile laying, alignment checking, grouting, quality inspection, and safety practices. This practical exposure improved my understanding of flooring installation and quality control.

4.10 Conclusion

Flooring is an important finishing activity that provides a durable, level, and aesthetically pleasing surface. The practical exposure during my internship enhanced my understanding of flooring execution, quality inspection, and site safety practices.

5. False Ceiling

A false ceiling is a secondary ceiling installed below the main structural ceiling using a metal framework and gypsum boards. It improves the appearance of the interior, conceals building services, and enhances thermal and acoustic comfort.

5.1 Purpose

- To provide an aesthetically pleasing ceiling finish.
- To conceal electrical conduits, HVAC ducts, and plumbing services.
- To improve thermal and sound insulation.
- To provide a level surface for lighting fixtures and other ceiling accessories.
- To provide a concealed channel for ducting within the ceiling void.

Materials Required



Tools and Equipment

- Measuring tape
- Laser level / spirit level
- Drill machine
- Screw gun
- Metal cutter
- Ladder
- Chalk line

Procedure

1. The ceiling layout was marked according to the approved drawings.
2. The ceiling level was established using a laser level or spirit level.
3. Perimeter channels were fixed along the walls at the specified height.
4. Hanger rods were fixed to the structural slab at the required spacing.
5. Main channels and cross channels were installed to form the ceiling framework.
6. The framework alignment and level were checked before fixing the gypsum boards.
7. Gypsum boards were fixed to the framework using drywall screws.
8. Joint tape and jointing compound were applied to the board joints to obtain a smooth surface.
9. The completed false ceiling was inspected before painting.

Site Observations

1. The ceiling layout and level were marked as per the approved drawings before installation.
2. GI channels were installed to form a rigid framework.
3. Gypsum boards were fixed using drywall screws at uniform spacing.
4. Openings for light fixtures, air-conditioning vents, and other services were provided according to the drawings.
5. Joint tape and jointing compound were applied to all board joints before painting.
6. Four types of tiles were used in the false ceiling: (a) mineral fibre, (b) calcium silicate, (c) metal tiles, and (d) antimicrobial tiles.

Site Images of Materials



Gypsum board cartons at site



Mineral fibre tile cartons



Metal ceiling tiles (Gyproc)

Quality Inspection

1. The ceiling level and alignment were checked.
2. The spacing of hangers and GI channels was verified according to the approved drawings.
3. Gypsum boards were inspected for proper fixing and damage.
4. Board joints and screw heads were checked to ensure a smooth finish before painting.

Safety Precautions

1. Workers wore PPE, including safety helmets, safety shoes, and gloves.
2. Ladders and scaffolding were inspected before use.
3. Electrical tools were checked before operation.
4. Materials were handled carefully to prevent damage and injury.
5. The work area was kept clean to avoid accidents

5.2 Site Images of Work in Progress



Fixing the ceiling framework



Workers installing perimeter channels



Completed false ceiling framework with services routed above

Learning Outcome

During my internship, I learned the installation sequence of false ceiling systems, including framework fixing, gypsum board installation, joint treatment, quality inspection, and safety practices.

Conclusion

False ceiling work improves the appearance and functionality of interior spaces by providing a level ceiling surface and concealing building services. The practical exposure during my internship enhanced my understanding of false ceiling installation, quality control, and safe construction practices.

6. Painting Work

Painting is the final finishing process in which primer and paint are applied to walls and ceilings to protect the surface and enhance the appearance of the building.

6.1 Purpose

- To protect the surface from moisture, dust, and weathering.
- To improve the appearance of the building.
- To provide a smooth, durable, and decorative finish.

6.2 Materials Used

- Wall putty
- Primer
- Interior emulsion paint
- Antibacterial paint (for ceilings and specified hospital areas)



Interior emulsion paint



Interior wall primer



Wall putty

6.3 Tools and Equipment

- Paint roller
- Paint brush
- Sanding paper
- Ladder



Putty application tool



Paint brush

6.4 Procedure

1. The plastered surface was inspected and cleaned to remove dust, dirt, and loose particles.
2. Surface imperfections, holes, and cracks were repaired using wall putty.
3. The putty surface was sanded to obtain a smooth and even finish.
4. One coat of primer was applied to improve paint adhesion.
5. After the primer dried, the surface was lightly sanded wherever required.
6. Two coats of interior emulsion paint were applied to the walls using rollers and brushes.
7. Antibacterial paint was applied to the ceilings and other specified hospital areas as per the project specifications to maintain hygiene and provide resistance to bacterial growth.
8. Adequate drying time was maintained between each coat.
9. The completed painted surfaces were inspected for colour uniformity, smoothness, and overall finish.

6.5 Site Observations

1. Painting work was started only after plastering and putty work had been completed.
2. One coat of primer was applied before the final paint coats.
3. Two coats of emulsion paint were applied to the walls to achieve a uniform finish.
4. Antibacterial paint was used on the ceilings and in designated hospital areas to meet hygiene and infection-control requirements.
5. Rollers were used for large wall and ceiling surfaces, while brushes were used for corners, edges, and difficult-to-reach locations.

6.6 Site Images



Applying emulsion paint with a roller



Painted room ready for inspection



Mixing and preparing putty on site

6.7 Quality Inspection

1. The painted surfaces were checked for uniform colour, smoothness, and proper coverage.
2. The surfaces were inspected for defects such as peeling, blistering, pinholes, cracks, and roller marks.
3. The adhesion of the paint and the quality of the finish were verified before approval.
4. Touch-up painting was carried out wherever defects were observed.

6.8 Safety Precautions

1. Workers wore PPE, including safety helmets, gloves, safety shoes, masks, and safety goggles.
2. Ladders and scaffolding were inspected before use.
3. Adequate ventilation was maintained during painting.
4. Paint containers were stored safely and handled according to the manufacturer's recommendations.
5. The work area was kept clean to prevent slips and other accidents.

6.9 Learning Outcome

During my internship, I gained practical knowledge of surface preparation, putty application, primer coating, emulsion painting, the application of antibacterial paint in hospital areas, quality inspection, and the safety practices followed during painting works.

6.10 Conclusion

Painting is an essential finishing activity that improves the appearance, durability, and hygiene of a building. During my internship, I learned the complete painting process and observed the use of antibacterial paint in hospital ceilings and other specified areas to meet healthcare facility standards.

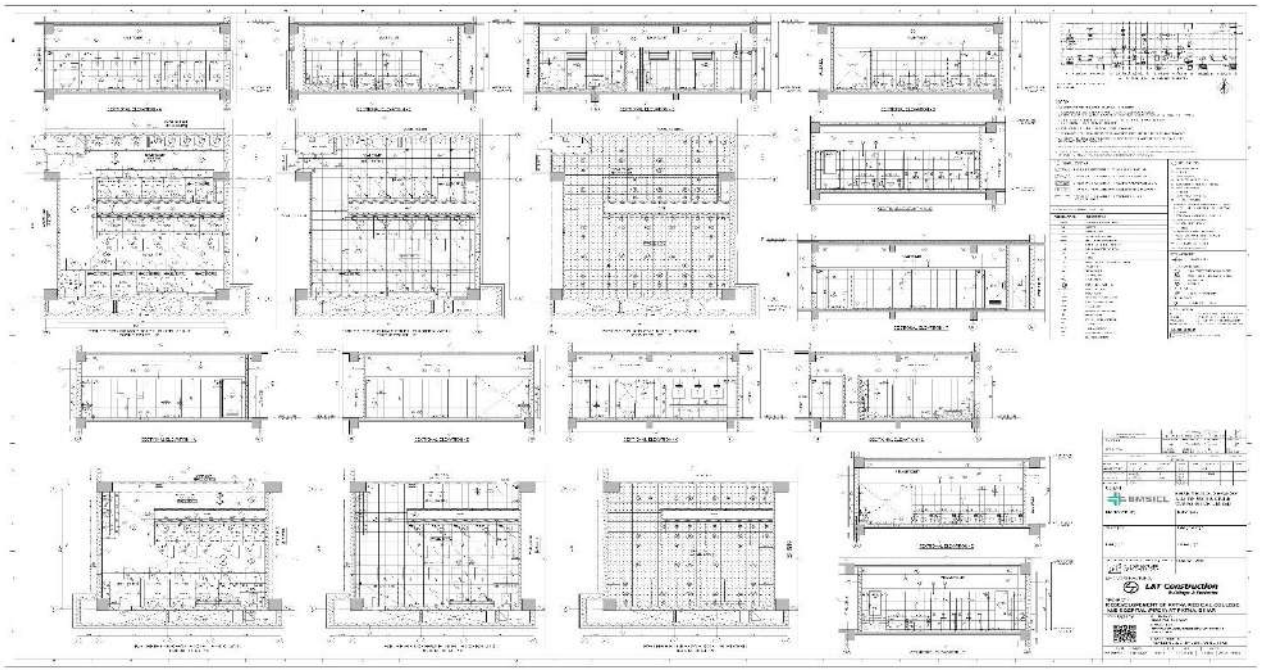
some Facilities

Hospital Toilet Facilities

During my internship at the PMCH Redevelopment Project, I observed the finishing works of the hospital toilet facilities. The toilet layout drawing was studied to understand the arrangement of the wash basins, urinals, toilet cubicles, doors, and other sanitary fixtures. By comparing the layout drawing with the completed work on site, I learned how the architectural design is implemented during construction.

During the site visit, I observed that ceramic wall and floor tiles had been completed, and the wash basin counter, mirrors, and urinals were installed according to the approved drawings. The toilet was also provided with a **false ceiling made of calcium silicate boards**, which provides a smooth finish, is moisture-resistant, and is suitable for humid environments such as hospital toilets.

Hospital toilet facilities are essential for maintaining hygiene, sanitation, and user comfort for patients, visitors, and hospital staff. Through this observation, I gained a better understanding of toilet finishing works and the importance of following architectural drawings during construction.



Layout Drawing of toilet

Site images



Wash basin outlets before fixture installation



Wash basin counter with mirrors installed



Urinals installed in the hospital toilet

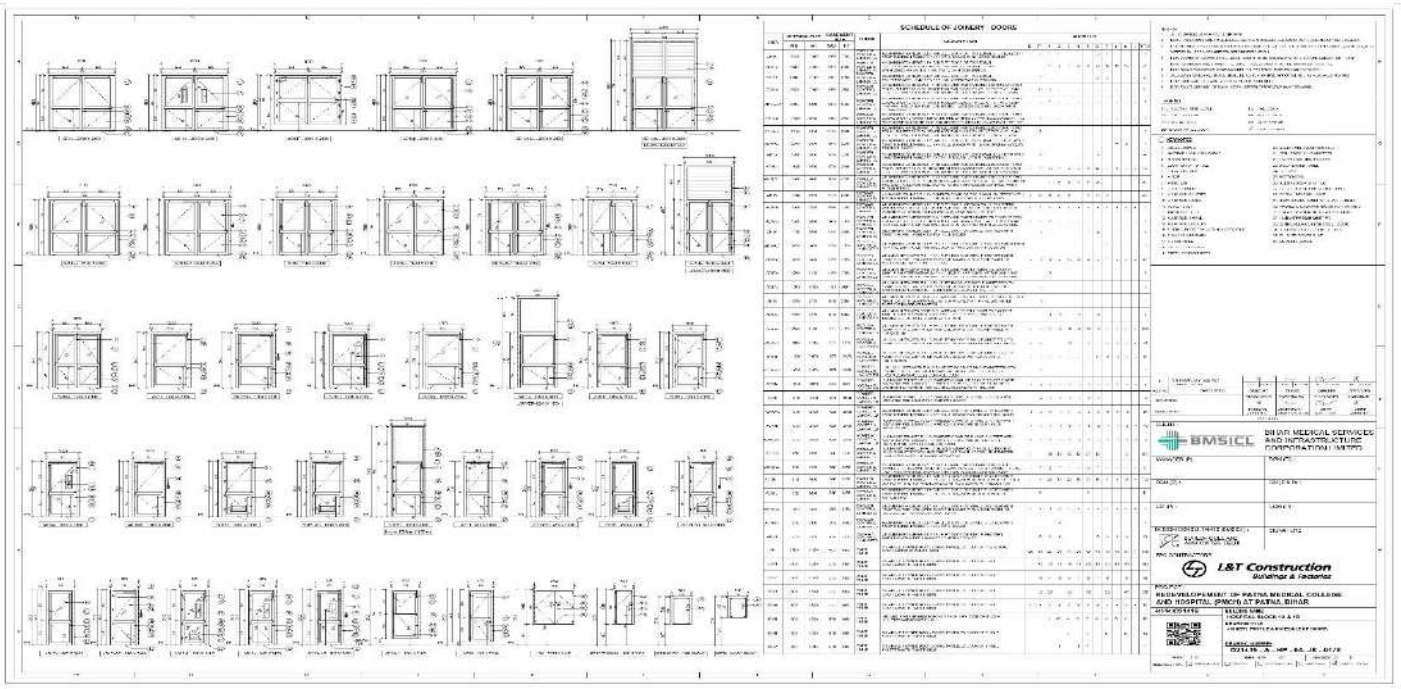
Hospital Doors

During my internship at the PMCH Redevelopment Project, I observed the installation of different types of hospital doors, including fire-rated doors and double-leaf swing doors. The doors were installed with frames, vision panels, handles, and door closers in accordance with the approved architectural drawings.

After the door frames were fixed, **silicone sealant** was applied around the edges of the frames to fill the gap between the frame and the wall. The silicone sealant provides a neat finish, seals small gaps, prevents the entry of dust and moisture, and improves the overall appearance of the installation. In the case of fire-rated doors, the joints around the frame should be sealed using the specified fire-rated sealing materials as required by the project specifications.

Hospital doors are essential for providing safe access, maintaining privacy, and ensuring smooth movement of patients, visitors, and medical staff. Fire-rated doors help prevent the spread of fire and smoke, while doors with vision panels improve visibility and enhance safety in hospital corridors.

Layout drawing of door



The material used to fill the gap between a **door frame is Silicone Sealant – Applied after installation to seal the joint and provide a smooth, waterproof finish. This is commonly used around hospital door frames.**



Some door images



Fire-rated double-leaf door with vision panels



Door frame with protective wrapping prior to cleaning



Fire-rated single door (red) with vision panel



Glass double doors at corridor entrance

7. Other Technologies Observed at Site

7.1 Lead Rubber Bearing (LRB)

A Lead Rubber Bearing (LRB) is a type of seismic base isolation device installed between the building foundation and the superstructure.

Purpose

- To improve the earthquake resistance of the building.
- To reduce structural vibrations during seismic events.
- To protect the building and its occupants from earthquake damage.
- To minimise damage to sensitive medical equipment in the hospital.



Lead Rubber Bearings installed at the foundation level



7.2 Tube Pneumatic System (PTS)

A Pneumatic Tube System (PTS) is an automated transport system that uses compressed air or vacuum to transfer carriers through a network of tubes within a building.

Purpose

- To transport blood samples, medicines, documents, and laboratory specimens quickly.
- To reduce manual transportation within the hospital.
- To improve operational efficiency and reduce turnaround time.
- To enhance infection control by minimising unnecessary movement.





Pneumatic Tube System stations installed at site

8. Building Services and Utility Systems Observed at Site

During my internship, I observed several building service systems installed to ensure the efficient operation of the hospital. These systems play an important role in providing comfort, safety, and uninterrupted services to patients and staff.

Chiller Plant

The chiller plant is the central air-conditioning system of the hospital. It produces chilled water, which is circulated through AHUs (Air Handling Units) and FCUs (Fan Coil Units) to maintain the required indoor temperature and humidity.



Cooling plant installed at the rooftop

Solar Water Heating System

Solar heating panels were installed to utilise solar energy for heating water. This system reduces electricity consumption and promotes sustainable energy use within the hospital.

Solar water heating panels installed on the rooftop



Quality Department

8.1 Introduction

The Quality Department is responsible for ensuring that all construction activities are carried out in accordance with the approved drawings, project specifications, and relevant Indian Standards (IS Codes). During my internship, I observed that quality inspections were conducted at every stage of construction, from structural works to finishing works, to ensure the required standards of safety, durability, and workmanship.

8.2 Quality Checks During Structural Works

- Verification of reinforcement size, spacing, lap length, and concrete cover before concreting.
- Inspection of formwork dimensions, alignment, and stability.
- Checking the grade and slump of concrete before placement.
- Collection of concrete cube samples for compressive strength testing.
- Monitoring proper concrete compaction and curing.
- Inspection of structural members after de-shuttering for dimensions, honeycombing, and surface defects.

8.3 Quality Checks During Finishing Works

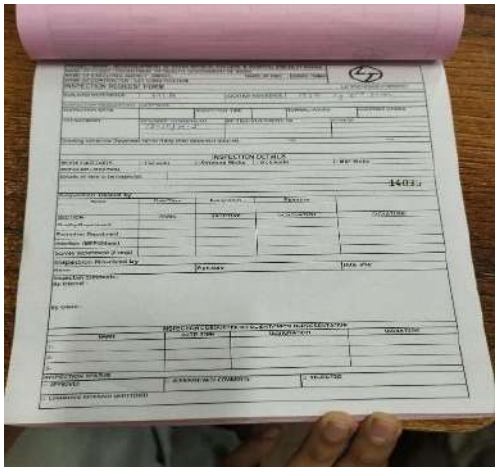
- Verification of work as per the approved drawings and specifications.
- Inspection of the alignment, level, and dimensions of finished work.
- Checking the quality of materials before use.
- Inspection of surface finish, workmanship, and overall appearance.
- Identification and rectification of defects before the next activity or final handover.
- Ensuring that all finishing works met the required quality standards before client inspection.

During my internship, I learned the importance of quality control in both structural and finishing works. I observed that regular inspections, testing, and adherence to project specifications helped ensure the successful completion of the project to the required quality standards.

8.4 Quality Records and Inspections



Concrete cube samples cast for compressive strength testing



Inspection Request (IR) form completed prior to commencing work

Site visit by the company's senior management to review the quality of ongoing work



Overall Safety Observed at Site

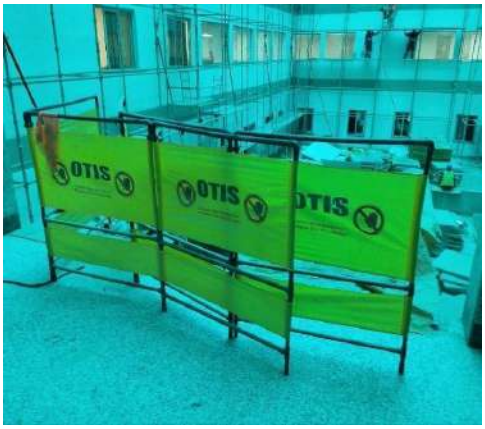
- All personnel entering the site were required to wear personal protective equipment (PPE), including safety helmets, reflective jackets, safety shoes, gloves, and safety goggles.
- A safety induction was conducted for workers and visitors before entering the construction site.
- Safety signboards and warning notices were displayed at different work locations.
- Scaffolding and ladders were inspected regularly before use.
- Fire extinguishers and first-aid kits were provided at suitable locations.
- Barricades and caution tapes were installed around hazardous areas.
- Housekeeping was maintained to keep walkways free from debris and construction waste.
- Daily toolbox talks were conducted to create awareness of site hazards and safe working practices.



Safety induction booth at the site entrance



Worker using a safety harness on scaffolding



Barricading around a hazardous work area



Safety poster on correct manual liking technique

Overall Conclusion

This internship provided valuable hands-on exposure to the finishing works involved in large-scale hospital construction, from brickwork and plastering through to flooring, false ceiling installation, and painting. Beyond the finishing trades, the internship also offered insight into specialised building technologies such as Lead Rubber Bearings and Pneumatic Tube Systems, as well as essential building services including chiller plants and solar water heating systems.

Throughout the internship, I gained a strong appreciation for the role of the Quality Department in enforcing adherence to approved drawings, specifications, and Indian Standards at every stage of construction, and for the rigorous safety culture maintained on site. This practical experience has significantly strengthened my understanding of construction execution, quality control, and site safety management, and has provided a solid foundation for my continued growth as a civil engineering professional.